



Engr. Hasnain Haroon

Civil Engineer | PEC Registered — CIVIL/081808

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Portfolio

PROFILE SUMMARY

PEC-registered Civil Engineer with a structural focus and research interest in AI-based structural health monitoring. Experienced in site supervision, quantity surveying, construction documentation, and AutoCAD drafting on an embassy project in Islamabad, progressing from Paid Intern to Civil Engineer, with structural and surveying internships. Developed a YOLOv11s-seg crack detection system as a final year project (2nd Position, Open House & Job Fair 2026). Seeking opportunities to contribute to infrastructure and construction projects while continuing professional growth.

EDUCATION

B.Sc. Civil Engineering — COMSATS University Islamabad, Wah Cantt Campus	2022 – 2026
CGPA: 3.77/4.00. Graduated July 2026. Specialized in structural engineering with active research in AI-based structural health monitoring. PEC-accredited Level 2 program under the Washington Accord.	
Intermediate (FSc Pre-Engineering) — Wah Hope College, Fateh Jang	2020 – 2022
Matriculation (Science) — Govt. Boys High School No. 2, Fateh Jang	2018 – 2020

EXPERIENCE

Civil Engineer — Huzaim Kamal Enterprises (HKE)	Jul 2026 – Present
<i>Construction of Afghanistan Embassy, Diplomatic Enclave, Islamabad</i>	
- Supervise and monitor civil construction; inspect reinforcement, formwork, and concrete works to ensure compliance with approved drawings, specifications, and quality requirements. - Perform quantity take-offs, site measurements, and verification of construction quantities and materials, assisting the Quantity Surveyor with BOQ and billing. - Prepare, review, and modify AutoCAD drawings; coordinate with senior staff, contractors, and consultants on documentation and site queries.	
Paid Intern — Huzaim Kamal Enterprises (HKE)	Sep 2025 – Jul 2026
<i>Same project; balanced professional responsibilities with B.Sc. Civil Engineering studies (completed July 2026).</i>	
- Assisted in day-to-day site supervision and verification of work against approved drawings. - Prepared AutoCAD drawings and revisions as required by site and project teams. - Carried out site measurements and quantity take-offs to support estimation and billing.	
Trainee Engineer — Structure Office, NESPAK (Pvt.) Ltd.	14 Jul – 22 Aug 2025
<i>National Engineering Services Pakistan</i>	
- Worked in the Structure Office of one of Pakistan's premier engineering consultancies on structural analysis and design documentation. - Assisted in reviewing structural drawings, reinforcement details, and design calculations under senior engineers. - Gained exposure to office engineering workflows, design standards, and consultancy documentation practices.	
Survey Camp — Field Training, COMSATS University Islamabad	5 Aug – 19 Aug 2024
<i>Wah Campus — Harnoi, Abbottabad</i>	
- Intensive field camp in real terrain covering topographic survey, leveling, traversing, and contouring. - Recorded field data, performed computations and adjustments, and plotted survey results as contour maps. - Worked in survey teams, developing practical skills in instrument setup, measurement accuracy, and reporting.	

EXPERIENCE (CONTINUED)

Trainee Engineer — Site Engineer, Winston Developers (Pvt.) Ltd.	Jul – Aug 2024
- Supervised construction activities on a multi-storey commercial building. - Monitored quality control (QA/QC) and checked work against approved drawings. - Assisted in material estimation and site coordination.	
Trainee Engineer — Site Engineer, Hashim Associates & Builders Pvt. Ltd.	Jun – Jul 2024
- Supervised construction activities on residential housing projects and prepared progress reports. - Checked reinforcement and inspected formwork prior to concrete pouring. - Gained hands-on site management experience across the construction cycle.	

KEY PROJECTS

StructInsight AI — Final Year Project (2nd Position, Open House & Job Fair 2026): YOLOv11s-seg system for crack detection, segmentation, crack-width estimation (pixel-to-mm calibration), and four-class severity grading (Hairline to Severe); deployed as a live web app (structinsightai.streamlit.app) with an Android field-app companion.	
Quantity Cost Estimation — Multi-Storey RC Building (Jun 2026): Material take-off for footings, columns, beams, slabs, and stairs; reinforcement scheduled via BBS with 20% wastage; estimated cost Rs 20.7M.	
RC School Building Structural Design (ACI 318-14 & ASCE 7): Manual design of a 3-storey school (120'x180' plot) covering slabs, beams, and columns, with AutoCAD/Revit detailing.	
Residential House Architectural Design (Jun 2026): 2-storey house, 25'x50' plot, Islamabad — plans, elevations, sections, and schedule of openings; plus a 3D massing study of a 10-Marla house in CAD.	
Combined Sewerage System — SWMM 5.2 (Wah Cantt): Dynamic wave simulation with 168 subcatchments and 11 outfalls, including continuity analysis.	
Sedimentation & Coagulation Tank Design (Faisal Hills, Taxila): Jar test established optimum alum dose of 2 mg/L; tank designed at 20 m x 54 m x 2 m per WASA standards.	
Primavera P6 Scheduling & ETABS Models: CPM scheduling with resource leveling for a residential house; RC building models with seismic analysis and design checks per ACI & ASCE.	

TECHNICAL SKILLS

Engineering Software: AutoCAD (Expert), MS Word (Expert), ETABS, Revit, Primavera P6, SWMM, RISA 2D, MS Excel (Intermediate), LaTeX (Beginner)
Structural & Civil: Site Supervision, Quantity Estimation, Construction Documentation, Engineering Drawing, Structural Analysis, RC & Steel Design, Foundation Design, Load Analysis, Surveying
AI & Programming: Python, YOLOv11 / Deep Learning, Computer Vision, Image Processing (Intermediate)
Professional Skills: Teamwork, Communication, Problem Solving, Time Management, Project Management, Technical Writing, Critical Thinking

LANGUAGES

Urdu — Native | English — Professional

CERTIFICATIONS & AWARDS

2nd Position — Open House & Job Fair, COMSATS Wah (Jun 2026) - Treasurer, ICE CUI Wah — Certificate of Appreciation (2025–26) - Civil Nexus 2025, ASCE University of Wah (Apr 2025) - Webinar — AI/ML in Civil Engineering, Pakistan Engineers Forum (Sep 2024) - Community Services, Community Organizer & Dengue/Malaria Awareness — CUI Wah (2024) - Workshop on Introduction to R — UNESCO Chair, COMSATS (Nov 2023) - Certificate of Appreciation — Pakistan Red Crescent Society (May 2023) - Truss Bridge Competition, ICE CUI Wah (Dec 2022)
CPD Seminars — CUI Wah Campus - Finite Element Method (May 2025) - Restoration of National Heritages (Feb 2025) - Rut Depth Modeling in Asphalt (Dec 2024) - NDT Methods for Concrete Bridge (May 2024) - Integrating Technologies in Civil Eng. (Sep 2023) - Machine Learning in Civil Eng. (Oct 2022)